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RESEARCH

Pulsed Nd:YAG versus continuous-wave laser photobiomodulation for orthodontic pain: a split-mouth RCT

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Abstract

Local pain following orthodontic tooth movement negatively affects patients' experience and compliance. Pulsed neodymium-doped yttrium-aluminum-garnet (Nd:YAG) laser, based on photobiomodulation therapy (PBMT), can effectively relieve this pain. This study applied pulsed Nd:YAG laser at different frequencies to orthodontic sites and used digital techniques to evaluate differences in pain relief after tooth movement, aiming to screen the optimal frequency, optimize laser parameters, and provide scientific evidence for the analgesic efficacy of pulsed Nd:YAG laser in orthodontics. 96 patients receiving fixed orthodontic appliances for the first time, who met all inclusion criteria and gave informed consent, were enrolled in this randomized, double-blind, controlled trial. Participants were randomly assigned to one of three pulsed Nd:YAG laser groups—low frequency (10 Hz), medium frequency (15 Hz), or high frequency (20 Hz)—or to a continuous-wave semiconductor laser group ($n=24$ each). Each participant further served as his or her own control: within each group, one randomly chosen side received active laser treatment while the contralateral side received sham treatment. A digital pressure algometer was used to measure pressure-pain thresholds (PPT) at the crown and the apex of the target teeth at 0 h, 24 h, 72 h, and 7 days after bracket placement, following intra-oral quantitative sensory testing (QST) protocols. Simultaneously, a visual analogue scale (VAS) and questionnaires were employed to record pain onset, peak pain time, pain disappearance time, and daily pain intensity. All data were analyzed using SPSS 27.0. 1. Intra-oral QST findings: Crown PPT on the treated side in the 10 Hz pulsed Nd:YAG group was significantly higher than that on the treated side in the continuous-wave semiconductor laser group ($P<0.05$).In the 15 Hz pulsed Nd:YAG group, both apical and crown PPT values on the treated side were significantly higher than the corresponding values in the semiconductor laser group ($P<0.05$). For all three pulsed Nd:YAG frequencies and the continuous-wave laser, PPT values on the treated side were significantly higher than those on the non-treated side ($P<0.05$). 2. VAS and questionnaire findings: VAS scores on the treated side of the 10 Hz group were lower than those of the semiconductor laser group on days 2 and 3 ($P<0.05$). VAS scores on the treated side of the 15 Hz group remained significantly lower than those of the semiconductor laser group throughout the observation period (days 1–6) ($P<0.05$). From day 1 to day 6, VAS scores on the treated side were significantly lower than on the non-treated side in all four groups ($P<0.05$). The 15 Hz group also exhibited significantly shorter pain disappearance time and overall pain duration compared with the semiconductor laser group ($P<0.05$). 1. Under the PBMT paradigm, both pulsed Nd:YAG and continuous-wave semiconductor lasers effectively alleviate pain and associated symptoms after orthodontic tooth movement. 2. When total energy is kept constant (60 J), 15 Hz pulsed Nd:YAG laser shows promising analgesic effects with better reduction in tooth sensitivity and shorter pain duration than 10–20 Hz. Frequency and analgesia are not linearly correlated. The 15 Hz, 150 μs, 0.5 W setting appears to be a potentially optimal parameter for orthodontic pain relief.

Keywords Orthodontic treatment · Orthodontic pain · Pulsed neodymium doped yttrium aluminum garnet laser · Photobiomodulation therapy

Extended author information available on the last page of the article

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Introduction

Pain is a common complaint during orthodontic treatment. When fixed appliances are first placed, most patients report spontaneous pain, occlusal tenderness, and hypersensitivity to hot or cold stimuli within 2 h; the discomfort usually peaks at 24–48 h. Although the intensity subsides over time, ≈ 25% of patients still experience local pain 7 days after arch-wire activation [1].

Current management of orthodontic pain follows a conservative, step-wise, non-invasive strategy. The first option is to select an appliance system that minimizes tissue irritation; the second is to combine treatment with psychological intervention [2]; the third is medication, which is limited by well-documented side-effects [3, 4]. Over the last decade, laser-based photobiomodulation therapy (PBMT) has been increasingly investigated for pain control.

PBMT employs non-ionising optical radiation to initiate photophysical, photochemical and photobiological events within tissues [5, 6]. The neodymium-doped yttrium-aluminum-garnet (Nd:YAG) laser emits at 1064 nm, delivering high peak power and deeper tissue penetration than shorter wavelengths [7–9]. Moreover, in recent years, the biological effects of Nd:YAG lasers have been widely applied in clinical settings and have achieved certain results [10, 11]. Nevertheless, data on Nd:YAG lasers for orthodontic pain are still scarce.

Therefore, the present study compared three pulsed Nd:YAG frequencies (low 10 Hz, medium 15 Hz, high 20 Hz) with an 808 nm continuous-wave semiconductor laser. Intra-oral pressure-pain threshold (PPT) and visual-analogue scale (VAS) scores were used to quantify pain relief under each laser condition. By means of within-subject and between-group comparisons we aimed to determine the differential efficacy of pulsed Nd:YAG versus semiconductor PBMT and to provide objective evidence for the clinical usefulness of frequency-varied Nd:YAG laser in reducing orthodontic pain.

Materials and methods

Experimental materials

(1) Damon 2 standard-torque self-ligating brackets (Ormco Corporation, USA).
(2) Nd:YAG laser therapy unit (Fotona, Slovenia).
(3) Semiconductor laser therapy unit (DoctorSmile, Italy).
(4) Digital pressure algometer (Wagner FPX, USA).

Study participants

Participants were recruited from the Department of Stomatology, Air Force Medical Center. A total of 96 individuals who met the inclusion criteria and volunteered to receive laser photobiomodulation for orthodontic pain relief were enrolled, including 12 minors aged 12–18 years. Written informed consent was obtained from all adults, and parental consent from minors. This study was conducted in accordance with the Declaration of Helsinki and approved by the Ethics Committee of the Air Force Medical Center (Approval No.: 2024-23-PJ01). Clinical Trial Registration: Chinese Clinical Trial Registry (ChiCTR, ChiCTR2500111300), registered on 29 October 2025. Patient recruitment commenced after trial registration, in accordance with prospective registration requirements.

Inclusion and exclusion criteria

Inclusion criteria

(1) Good general health, free of systemic diseases.
(2) No unilateral chewing habit; upper and lower dental arches essentially symmetrical.
(3) Permanent dentition with complete dentition.
(4) Crowding as the main occlusal anomaly.
(5) No previous orthodontic treatment.
(6) No use of non-steroidal anti-inflammatory drugs within the past month.

Exclusion criteria

(1) Impacted, ectopic or high-positioned maxillary cusp(s).
(2) Functional disorders, e.g., temporomandibular disorders.
(3) Presence of pulpitis or periodontitis.

Experimental protocol

Group allocation

This study employed a randomized, controlled, double-blind, split-mouth design. Blinding was implemented as follows: participants were blinded to group allocation and treatment type (active/sham); Operator 2, who performed all outcome measurements, was blinded to group assignment and treatment side, while Operator 1, responsible for randomization and laser delivery, was unblinded but not involved in outcome assessment. Operator 1 performed randomization and delivered all laser interventions; Operator 2, blinded to group allocation, carried out all outcome measurements. This study used a 2×2 hybrid

Table 1 PBMT parameter settings

Group	Wavelength	Power	Emission mode	Pulse width	Irradiation distance	Irradiation time	Energy	Spot diameter	Irradiated area	Fluence
Nd:YAG Laser	1064 nm	0.5 W	Pulse Mode	150μs	1 cm	2 min	60 J	1 mm*	0.0079 cm^2	7.6×10^3 J/cm^2
Semiconductor laser	808 nm	0.5 W	Continuous-wave (CW) mode	-	1 cm	2 min	60 J	1 mm*	0.0079 cm^2	7.6×10^3 J/cm^2

*The spot diameter was measured using infrared photosensitive paper at a working distance of 1 cm, with the width taken as $1/e^2$

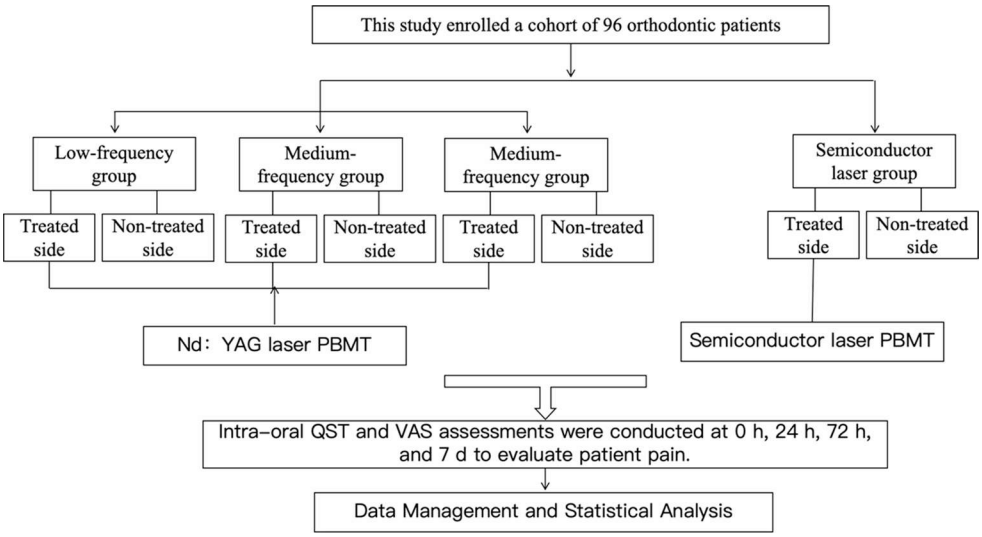


Fig. 1 Technology roadmap

design: a between-subject four-arm comparison (three pulsed Nd:YAG laser frequencies vs. one continuous-wave semiconductor laser) combined with a within-subject split-mouth design (treated side vs. sham-treated side in each patient).

All three pulsed Nd:YAG arms received PBMT with the Nd:YAG laser: Low-frequency group (10 Hz), Medium-frequency group (15 Hz), High-frequency group (20 Hz). All experimental arms received photobiomodulation therapy (PBMT) with the Nd:YAG laser. The semiconductor laser arm received PBMT with a continuous-wave semiconductor laser; laser parameters were detailed in Table 1. Within each of the four arms, computer-generated randomization further assigned each participant a "treated side" and a "non-treated side" (split-mouth). Thus, inter-group comparisons were conducted between the three Nd:YAG groups and the semiconductor laser group, while intra-group comparisons were made between the treated and non-treated sides within each participant.

Technology roadmap (Fig. 1)

Photobiomodulation therapy

The experimental groups received pulsed Nd:YAG laser (λ = 1064 nm) at frequencies of 10 Hz, 15–20 Hz according to allocation; the semiconductor laser group was treated with a continuous-wave semiconductor laser (λ = 808 nm).

Laser application began immediately after bonding Damon 2 standard-torque self-ligating brackets and inserting a 0.013-inch CuNiTi arch-wire. Both operator and participant wore protective eyewear; the participant was instructed to keep the eyes closed throughout the procedure.

On the "treated" side, eight sites on the upper canine were irradiated: mesial, distal, mid-root and apical aspects on both buccal and palatal surfaces. Each site received 15 s of exposure, giving a total irradiation time of 2 min. The hand-piece was held perpendicular to the long axis of the tooth at a distance of 1 cm and kept motionless.

On the "non-treated" side, the hand-piece was identically positioned over the same eight sites for the same duration; the pilot lamp was illuminated to mimic active treatment, but no laser energy was emitted (sham irradiation). See Fig. 2.

Outcome measures

The primary outcome was pressure pain threshold (PPT) measured by intra oral quantitative sensory testing (QST).

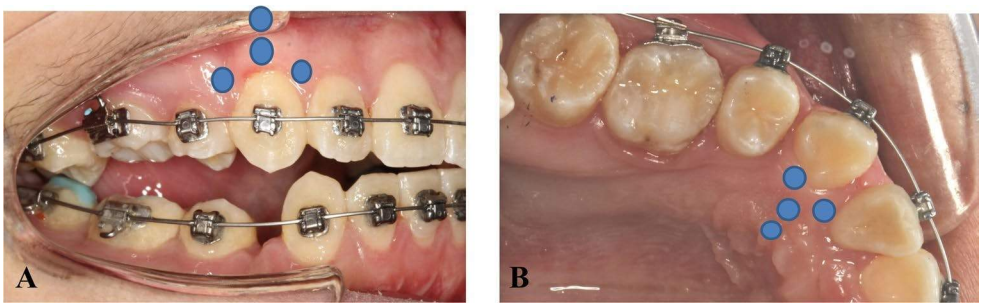


Fig. 2 Laser irradiation site. **A** The laser irradiation site on the buccal side of the tooth. **B** The laser irradiation site on the palatal side of the tooth

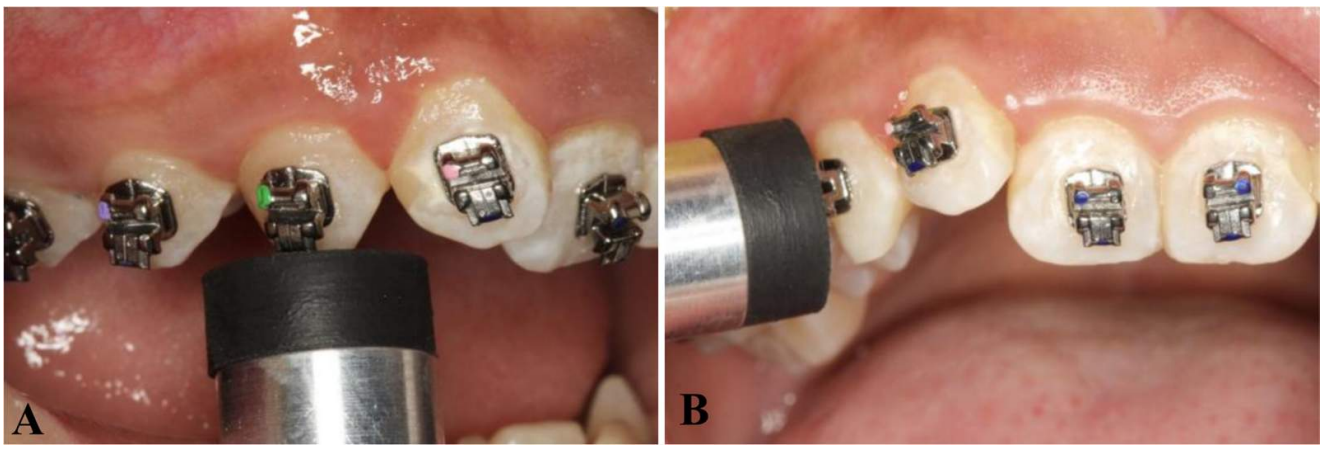


Fig. 3 PPT measurement of the tooth. **A** PPT force-recording site at the cusp tip. **B** PPT force-recording site on the crown surface

Fig. 4 Visual analogue scale



Secondary outcomes included visual analogue scale (VAS) pain scores, pain onset time, peak pain time, pain disappearance time, and total pain duration.

Intra-oral quantitative sensory testing – PPT

PPT was determined with the digital pressure algometer. Before each measurement session, the arch-wire was removed from the brackets. The algometer was held perpendicular to the tooth surface (Fig. 3) and pressure was increased at a constant rate until the participant felt the first sensation of pain and raised a hand. The force recorded at that moment (in pounds-force, lbf) was taken as the PPT. To ensure reliability, three consecutive readings were obtained at each site and averaged.

The sequence of recordings was: tip of the maxillary left canine, tip of the maxillary right canine, buccal surface of the crown of the maxillary left canine, buccal surface of the crown of the maxillary right canine. Values obtained at the cusp tip were designated "cusp PPT"; those obtained on the buccal crown surface were designated "crown PPT". Based on previously reported pain dynamics [2], PPT was measured at 0 h, 24 h, 72 h, and 7 days after bracket bonding.

VAS and questionnaire

During the study week following initial arch-wire placement, participants completed a questionnaire assessing post-operative pain. Items were as follows:

(1) When did pain on the left side first appear?
(2) When did pain on the right side first appear?
(3) When was pain on the left side most severe?
(4) When was pain on the right side most severe?
(5) When did pain on the left side disappear?
(6) When did pain on the right side disappear?
(7) Daily pain score for the left side over the 7-day period.
(8) Daily pain score for the right side over the 7-day period.

Items 7 and 8 were recorded using a 100-mm VAS (Fig. 4).

Statistical analysis

All data were analyzed using SPSS 27.0. Between-group comparisons were performed using one-way ANOVA with LSD post hoc tests. Within-subject paired comparisons between treated and non-treated sides were analyzed using paired samples t-tests. Repeated measures ANOVA was

Table 2 Age characteristics of the four study groups

Group	Age	*F*	*P*
Low-frequency group	20.73 ± 5.23	1.23	0.46
Medium-frequency group	19.94 ± 4.67		
High-frequency group	21.09 ± 5.11		
Semiconductor laser group	20.48 ± 4.26		

Table 3 Gender distribution across the four study groups

Group	Male	Female	$\chi 2$	*P*
Low-frequency group	10(41.7%)	14(58.3%)	0.83	0.84
Medium-frequency group	11(45.8%)	13(54.2%)		
High-frequency group	12(50%)	12(50%)		
Semiconductor laser group	13(54.2%)	11(45.8%)		

Table 4 Comparison of mean PPT values on the treated side

Group	Cusp PPT (lbf)	Crown PPT (lbf)
Low-frequency group	7.32 ± 1.28	2.89 ± 0.67*
Medium-frequency group	7.72 ± 1.44*	2.93 ± 0.42*
High-frequency group	7.25 ± 1.32	2.73 ± 0.59
Semiconductor laser group	6.91 ± 0.82	2.61 ± 0.55
F	3.39	4.66
P	<0.05	<0.05

*denotes statistically significant difference from the semiconductor laser group by LSD post-hoc test

used to evaluate time-dependent changes in PPT and VAS scores. The significance level was set at $\alpha = 0.05$.

Results

General characteristics ($n = 24$)

No statistically significant differences were observed in gender or age among the four groups ($P > 0.05$) (Tables 2 and 3).

Intra-oral QST – PPT

Inter-group comparison of mean PPT on the treated side

Mean cusp and crown PPT values at 0 h, 24 h, 72 h, and 7 days differed significantly across the four interventions ($P < 0.05$). LSD post-hoc tests showed that the medium-frequency (15 Hz) group exhibited significantly higher cusp and crown PPT values on the treated side than the semiconductor laser group ($P < 0.05$); the low-frequency (10 Hz) group also demonstrated a significantly higher crown PPT on the treated side compared with the semiconductor laser group ($P < 0.05$) (Table 4).

Within-group comparison (treated vs. non-treated side)

In every group, PPT values on the treated side were significantly higher than those on the non-treated side ($P < 0.05$) (Table 5).

Comparison of percentage change in cusp and crown PPT on the treated side

Among the four groups, the medium-frequency (15 Hz) pulsed Nd:YAG group showed the greatest percentage change in cusp PPT on the treated side. The curve dropped to its lowest point at 24 h (0.69), rose to 0.88 at 72 h, and reached 0.95 by day 7; at each of these time-points the medium-frequency group's values exceeded those of the other groups.

LSD post-hoc pairwise comparisons revealed that the percentage changes in both cusp and crown PPT on the treated side in the medium-frequency group were significantly higher than in the semiconductor laser group at 24 h and 72 h ($P < 0.05$). These findings indicate that the medium-frequency protocol produced a marked early- and mid-phase improvement in cusp and crown PPT on the treated side, demonstrating a superior analgesic effect and a greater elevation of pressure-pain thresholds during orthodontic tooth movement (Table 6).

VAS and questionnaire

Daily VAS scores on the treated side

VAS scores on the treated side in all four groups followed a characteristic pattern: a rapid rise on the first day, then a gradual decline, approaching 0 on day 6 and reaching zero (no pain) on day 7.

Post-hoc LSD pairwise comparisons showed that the low-frequency (10 Hz) group had lower VAS scores than the semiconductor laser group on days 2 and 3 ($P < 0.05$), whereas the medium-frequency (15 Hz) group exhibited significantly lower VAS scores than the semiconductor laser

Table 5 Comparison of mean cusp and crown PPT values between treated and non-treated sides

* indicates a statistically significant difference between the treated and non-treated sides

Group	Cusp PPT (lbf)		Crown PPT (lbf)	
	Treated side	Non-treated side	Treated side	Non-treated side
Low-frequency group	7.32 ± 1.28*	6.12 ± 1.09	2.89 ± 0.67*	2.29 ± 0.35
Medium-frequency group	7.72 ± 1.44*	6.19 ± 1.12	2.93 ± 0.42*	2.31 ± 0.44
High-frequency group	7.25 ± 1.32*	6.23 ± 1.17	2.73 ± 0.59*	2.18 ± 0.46
Semiconductor laser group	6.91 ± 0.82*	6.02 ± 0.92	2.61 ± 0.55*	2.02 ± 0.42

Table 6 Percentage change in cusp and crown PPT on the treated side

Group	Cusp PPT				Crown PPT			
	0 h	24 h	72 h	7d	0 h	24 h	72 h	7d
Low-frequency group	1	0.63	0.83	0.91	1	0.7	0.77	0.89
Medium-frequency group	1	0.69*	0.88*	0.95	1	0.73*	0.87*	0.93
High-frequency group	1	0.59	0.79	0.87	1	0.64	0.72	0.84
Semiconductor laser group	1	0.52	0.72	0.83	1	0.59	0.68	0.8

* denotes statistically significant difference from the semiconductor laser group by LSD post-hoc test

Table 7 Comparison of daily VAS scores on the treated side

Group	1d	2d	3d	4d	5d	6d	7d
Low-frequency group	2.91 ± 1.05	3.26 ± 0.97*	2.68 ± 1.32*	1.65 ± 1.03	0.92 ± 0.79	0.59 ± 0.53	0
Medium-frequency group	2.17 ± 1.14*	2.93 ± 1.39*	2.12 ± 0.86*	0.92 ± 0.65*	0.53 ± 0.64*	0.26 ± 0.33*	0
High-frequency group	2.99 ± 1.43	3.66 ± 1.56	3.13 ± 1.19	1.97 ± 1.23	1.12 ± 1.11	0.74 ± 0.58	0
Semiconductor laser group	3.27 ± 1.31	4.28 ± 1.72	3.56 ± 1.62	2.24 ± 1.64	1.33 ± 1.05	0.79 ± 1.17	0
F	6.29	5.52	7.13	5.46	4.99	3.05	
P	<0.05	<0.05	<0.05	<0.05	<0.05	<0.05	

* denotes statistically significant difference from the semiconductor laser group by LSD post-hoc test

group on every day from day 1 to day 6 ($P<0.05$). These results indicate that the low-frequency protocol can reduce orthodontic pain at certain time-points, but the medium-frequency protocol provides consistently better analgesic efficacy (Table 7).

Daily VAS comparison between treated and non-treated sides

Across all four groups, the daily VAS scores on the treated side were significantly lower than those on the non-treated side during the first six days ($P<0.05$). This indicates that, regardless of the specific protocol employed, laser application effectively alleviated orthodontic pain. Because all treated sides reported a VAS of 0 on day 7, no statistical comparison was performed for that time point (Table 8).

Pain-duration comparison among treated sides

Compared with the semiconductor laser group, the medium-frequency (15 Hz) group exhibited significantly shorter pain-offset times and overall pain durations on the treated side ($P<0.05$). This indicates that the 15 Hz protocol not only accelerates pain resolution but also markedly reduces the total period of discomfort (Table 9).

Pain-timing comparison between treated and non-treated sides

Within each group, laser application to the treated side significantly shortened total pain duration versus the non-treated side ($P<0.05$). However, neither pain onset time nor peak-pain time differed significantly between the two sides (Table 10).

Discussion

Optimisation of PBMT parameters for Nd:YAG and semiconductor lasers

Grounded in the unique photobiological effects of the Nd:YAG laser and informed by two prior clinical trials from our group on pulsed-Nd:YAG alleviation of orofacial pain [12, 13], we integrated findings from Akiko Sekine [14], Qiu Ni [15], Carrol [16], Zhao Jizhi [17], and others to define the following laser settings: wavelength 1064 nm, pulse width 150 μs, power density 0.5 W, and frequency gradients of 10, 15 and 20 Hz. Parameter selection balanced established literature with the deliberate introduction of a frequency gradient, thereby expanding the optimisation envelope for future clinical protocols.

Nd:YAG laser photobiological effects on orthodontic teeth and orthodontic pain

We used QST to provide an objective appraisal of orthodontic pain. QST is a validated tool for dissecting the neurobiological underpinnings of orthodontic discomfort [18]. PPT, a QST subset, evokes C- and Aδ-fibre activation by digital algometry; nociceptive signals ascend via the spinothalamic tract to thalamic and higher centres [19]. Local PPT recordings at cusp tips and crown buccal surfaces therefore mirror periodontal ligament innervation density and indirectly reflect ligament sensitivity [20].

Across both experimental and semiconductor laser groups, treated sides exhibited significantly higher cusp and crown PPT values than non-treated sides (Table 5), indicating that Nd:YAG and semiconductor PBMT confer comparable analgesic efficacy and enhance pressure tolerance of

Table 8 Daily VAS comparison between treated and non-treated sides

Group	1d		2d		3d		4d		5d		6d	
	Treated side	Non-treated side	Treated side	Non-treated side	Treated side	Non-treated side	Treated side	Non-treated side	Treated side	Non-treated side	Treated side	Non-treated side
Low-frequency group	2.91±1.05*	4.76±1.86	3.49±0.97*	7.36±1.74	2.68±1.32*	5.91±1.59	1.15±1.03*	4.97±1.52	0.82±0.79*	3.06±1.31	0.39±0.53*	1.39±1.29
Medium-frequency group	2.17±1.14*	4.55±1.37	2.93±1.39*	7.13±1.67	2.12±0.86*	5.79±1.31	0.92±0.65*	4.54±1.62	0.53±0.64*	2.81±1.17	0.26±0.33*	1.23±1.15
High-frequency group	2.99±1.43*	4.87±2.33	3.66±1.56*	7.68±2.95	3.13±1.19*	6.13±1.78	1.97±1.23*	5.33±1.49	1.12±1.11*	3.26±1.18	0.74±0.58*	1.44±1.39
Semiconductor laser group	3.27±1.31*	4.96±1.94	4.28±1.72*	8.42±3.31	3.56±1.62	6.36±2.25	2.24±1.64*	5.74±1.55	1.33±1.05*	3.39±2.09	0.79±1.17*	1.62±1.96

* indicates a statistically significant difference between the treated and non-treated sides on the same day

orthodontically loaded teeth—findings aligned with Wu et al. [21].

When Nd:YAG frequencies were compared, only the 15 Hz protocol yielded significantly higher cusp and crown PPT values—and greater percentage PPT increases—than the semiconductor laser group (Table 6). Raising the frequency to 20 Hz abolished this advantage. These data confirm that "more is not necessarily better": an optimal pulse frequency (here 15 Hz) outperforms both lower and higher settings, corroborating our 2021 observation that the optimal parameter set was not the one with the highest pulse frequency [13].

To further capture patients' subjective perception of orthodontic pain and related pain dynamics, we supplemented PPT measurements with VAS ratings and structured questionnaires. In the present study, the VAS trajectories for the treated sides of both the experimental and semiconductor laser groups followed a nearly identical pattern: pain peaked at 24 h post-activation, then steadily declined, reaching its nadir on day 6 or 7. This temporal profile is consistent with previous reports employing semiconductor-laser PBMT [22–25], underscoring the broad analgesic efficacy of photobiomodulation in alleviating orthodontic pain.

Further between-group comparisons revealed that the 10 Hz low-frequency group achieved significantly lower VAS scores than the semiconductor laser group only on days 2 and 3 ($P<0.05$); no differences were observed on day 1 or from days 4 to 6. When the pulse frequency was raised to 15 Hz, the treated side showed significantly lower VAS values than the semiconductor laser group on every day from day 1 to day 6 ($P<0.05$). However, increasing the frequency to 20 Hz produced VAS values that, although numerically lower, no longer differed statistically from the semiconductor laser group (Table 7).

This frequency-dependent pattern suggests that the 10 Hz protocol exerts its PBMT effect predominantly in the first three days, a finding consistent with the reports of Eslamian et al. [26] and Fujiyama et al. [27], both of whom observed significant pain reduction within the initial 72 h after PBMT.

In the present study, both the experimental and semiconductor laser groups exhibited significantly shorter pain duration and earlier pain disappearance on the treated side compared with the non-treated side (Table 10). This aligns with the VAS data showing that all treated sides had reached 0 by day 7, whereas 18 participants still reported pain on the non-treated side—an observation that mirrors earlier clinical investigations [28, 29]. No inter-side differences were detected for pain onset or peak-pain time (Table 10). Previous work has linked orthodontic pain to inflammatory mediators released within the periodontal ligament after tooth movement, such as substance P (SP) [30, 31], interleukin-1β (IL-1β) [32], and prostaglandin E2 (PGE2) [33, 34]. Studies

Table 9 Comparison of pain timing on the treated side

Group	Pain onset (h)	Pain peak (d)	Pain offset (d)	Pain duration (d)
Low-frequency group	2.56 ± 1.52	2.16 ± 1.04	4.02 ± 1.33	3.91 ± 1.43
Medium-frequency group	2.77 ± 1.43	2.05 ± 0.73	3.63 ± 0.69*	3.51 ± 0.86*
High-frequency group	2.57 ± 1.26	2.31 ± 0.95	4.17 ± 1.42	4.06 ± 1.28
Semiconductor laser group	2.64 ± 1.17	2.46 ± 1.09	4.66 ± 0.84	4.55 ± 1.21
F	0.94	1.59	3.17	2.88
P	0.52	0.41	<0.05	<0.05

* denotes a statistically significant difference from the semiconductor laser group by LSD post-hoc test

Table 10 Comparison of pain timing between treated and non-treated sides

Group	Pain onset (h)		Pain offset (d)		Pain peak (d)		Pain duration (d)	
	Treated side	Non-treated side	Treated side	Non-treated side	Treated side	Non-treated side	Treated side	Non-treated side
Low-frequency group	2.56 ± 1.52	2.63 ± 1.01	4.02 ± 1.33*	5.94 ± 1.17	2.16 ± 1.04	2.13 ± 0.96	3.91 ± 1.43*	5.83 ± 1.05
Medium-frequency group	2.77 ± 1.43	2.68 ± 1.26	3.63 ± 0.69*	5.72 ± 1.33	2.05 ± 0.73	2.23 ± 1.13	3.51 ± 0.86*	5.61 ± 1.21
High-frequency group	2.57 ± 1.26	2.34 ± 0.93	4.17 ± 1.42*	6.04 ± 0.89	2.31 ± 0.95	2.36 ± 1.27	4.06 ± 1.28*	5.94 ± 0.86
Semiconductor laser group	2.64 ± 1.17	2.45 ± 1.14	4.66 ± 0.84*	6.41 ± 0.62	2.46 ± 1.09	2.52 ± 1.11	4.55 ± 1.21*	6.31 ± 1.31

*indicates a statistically significant difference between the treated and non-treated sides

indicate that PBMT suppresses gingival crevicular levels of IL-1β, PGE2, and SP only after 24 h of irradiation [35], a time course that corresponds closely to the onset and peak pain times observed here.

Collectively, these findings suggest that, at an optimal frequency (15 Hz), pulsed Nd:YAG laser outperforms semiconductor laser in reducing dental hypersensitivity, alleviating orthodontic pain, and shortening pain duration. The key differences between the two systems are wavelength (1064 nm vs. 808 nm) and emission mode (pulsed vs. continuous-wave). Pulse frequency is a critical determinant of PBMT efficacy for the following reasons:

(1) Relationship between pulse frequency, penetration depth, and tissue heating

Nd:YAG laser is a pulsed system that rapidly stores and releases energy within each pulse by precisely modulating parameters such as pulse width, energy, and frequency. Consequently, the number of photons emitted per unit time is governed by the combined values of pulse width and frequency. Because the pulse duration can be adjusted to be shorter than the thermal relaxation time (TRT)—the interval required for the target tissue to cool to 50% of its peak temperature—heat diffusion to surrounding healthy tissues is intrinsically limited. This self-limiting thermal behaviour markedly reduces the risk of collateral thermal damage [36–38].

(2) Relationship between pulse frequency and therapeutic efficacy

Photobiomodulation delivered by pulsed lasers enhances the interaction between light and biological systems, thereby amplifying target-tissue responsiveness. Pulse frequencies in the tens-to-hundreds-of-hertz range approximate the intrinsic oscillatory frequencies of biological tissues, enabling pulsed lasers to interact more effectively with their targets than continuous-wave devices [39]. In the present study, the 1 064 nm pulsed Nd:YAG laser yielded superior clinical outcomes under PBMT, possibly because the chosen pulse frequency influences both tissue penetration and cellular metabolism [40, 41]; the precise mechanisms warrant further investigation.

Limitations and strengths of this study

One limitation of this study lies in the heterogeneity of the research sample, which included individuals of varying ages and genders, encompassing adolescents, adults, males, and females. Although considerable debate persists in the academic community regarding how age and gender influence the perception of orthodontic pain [42–44], the gender and age distributions across the four intervention groups in this study did not exhibit statistically significant differences. This finding provides a degree of support for the reliability of the research results. A second limitation of this study is that VAS, a commonly used method for assessing orthodontic pain, is highly subjective and may compromise the accuracy of evaluation results. Therefore, this study incorporated the application of PPT as an objective, visual indicator to quantitatively assess changes in subjects' tooth sensitivity to pressure following orthodontic treatment. Through the combined use of VAS and QST, this study enabled a more comprehensive evaluation and analysis of orthodontic pain and

its progression among participants. Additionally, insufficient sample size represented another limitation of this study, potentially restricting the statistical power and generalizability of the findings.

In summary, under the total energy of 60 J used in the present study, both pulsed Nd:YAG laser and continuous-wave semiconductor laser can effectively relieve pain after orthodontic tooth movement. The 15 Hz pulsed Nd:YAG laser shows favorable and promising analgesic effects, with better improvement in tooth sensitivity and shorter pain duration compared with 10 Hz and 20 Hz pulsed Nd:YAG laser and continuous-wave semiconductor laser. The analgesic effect did not increase linearly with increasing of pulse frequency. This parameter setting may be a potentially optimal option for relieving orthodontic pain in clinical practice, and further verification with larger samples and multi-center studies is still needed.

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Data availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

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